Tracing the complexity profiles of different linguistic phenomena through the intrinsic dimension of LLM representations

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Abstract

We explore intrinsic dimension (ID) of LLM representations as a marker of linguistic complexity. Specifically, we test whether ID differences across model layers reflect well-known complexity contrasts established in (psycho)linguistics: coordination vs. subordination, right-branching vs. center-embedding, and unambiguous vs. ambiguous attachment. Our results on six different LLMs show that these contrasts are consistently reflected in ID differences, with more complex phenomena eliciting higher ID profiles. Notably, ID differences emerge at different points across layers for different contrasts, also reaching their peaks at different stages. Further experiments using representational similarity and layer pruning confirm the trends. We conclude that ID is a useful marker of linguistic complexity in LLMs, that it points to similar linguistic processing steps across disparate LLMs, and that it has the potential to differentiate between different types of complexity.

1 Introduction

Given LLMs' remarkable linguistic skills, there is widespread interest in understanding how language processing unfolds in their inner layers, both because a better understanding of their inner workings could lead to more efficient and controllable models (Balestriero et al., 2024; Gromov et al., 2025), and because LLMs are imperfect but practical in silico models of the human faculty of language (see Futrell and Mahowald, 2025; Ziv et al., 2026 for contrasting views on their usefulness as cognitive models).

In this context, the *Intrinsic Dimension* (ID) of LLMs' inner representations has been used to gather insights on their internal processing stages (e.g., Valeriani et al., 2023; Doimo et al., 2024; Cheng et al., 2025). When datapoints are mapped to high-dimensional representations (e.g., sentences embedded in an LLM layer space), i) they will typically lie near a manifold that effectively occupies a much smaller number of dimensions than the full space, i.e., the intrinsic dimension is much lower than the so-called ambient dimension; and ii) the more complex the data are for the representational system (in our case, the LLM layer), the higher this intrinsic dimension will be. Consequently, we can use ID as a probe for complexity of linguistic processing across LLM layers.

It has been shown that LLMs have consistent ID profiles across their layers, and that these profiles cue phases of more or less abstract linguistic processing. Earlier studies, however, only looked at average model behaviour in response to generic corpus data (Cai et al., 2021; Cheng et al., 2025) or at very coarse distinctions between data types (Tulchinskii et al., 2023; Yin et al., 2024; Lee et al., 2025). Here, we take a more granular view and use ID to characterize how LLMs handle highly controlled complexity contrasts that have long been acknowledged in the (psycho)linguistic literature. In particular, we study, in English, i) the distinction between sentences with subordinated vs. coordinated clauses; ii) the contrast between center-embedding structures with long-distance agreement and equivalent sentences with right branching; and, finally, iii) sentences with attachment ambiguities compared to sentences where meaning disambiguates the attachment site.

We find a remarkable degree of consistency in how different LLMs process data characterized by these phenomena, and that their ID profiles point at different processing strategies for each of them. Our evidence from ID is further supported by experiments tracking representational similarities and pruning effects. Overall, our results suggest that sufficiently powerful systems converge to similar ways to handle language, even when they lack explicit priors for it, and consequently, that LLMs can provide us with new insights on the nature of linguistic complexity that can, in turn, complement

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human sentence processing research.

2 Related work

Intrinsic dimension While naturalistic language data appear high-dimensional, the *manifold hypothesis* posits they actually lie near a low-dimensional manifold (Goodfellow et al., 2016). The *intrinsic dimension* of the data is then the dimension of this possibly nonlinear manifold, that is, the number of degrees of freedom that explain it under minimal information loss (Campadelli et al., 2015). The manifold hypothesis holds not only for LLM parameter spaces (Aghajanyan et al., 2021; Zhang et al., 2023), but also for their activations: no matter the model or dataset, existing work shows that LLM representation manifolds have an ID orders-of-magnitude lower than their ambient dimension (Cai et al., 2021; Valeriani et al., 2023; Cheng et al., 2025). Similar to our work, Lee et al. (2025) and Cheng et al. (2023) show that representational ID over layers correlates to formal or psycholinguistic notions of linguistic complexity, in particular, n-gram diversity (Lee et al., 2025) and surprisal and learnability (Cheng et al., 2023). We build directly on the work by Cheng et al. (2025) who found, across different LLMs, a characteristic per-layer ID profile marked by an ID peak in intermediate layers. Through probing and downstream tasks, they show that this ID peak coincides with the phase where the model is first able to perform complex linguistic tasks, suggesting that the peak cues a stage of deep linguistic processing.

Linguistic probing of LLMs A rich literature has probed the internal representations of linguistic structures in LLMs, demonstrating that they encode aspects of syntactic knowledge and provide insights into where and how this information is stored (Belinkov and Glass, 2019; Linzen and Baroni, 2021; Rogers et al., 2020; Ferrando et al., 2024; Li and Subramani, 2025; Simon et al., 2025). Early work by Hewitt and Manning (2019) introduced a method based on linear probing, showing that bidirectional models represent information about syntactic dependencies and the relation between the involved constituents. Further research on more recent models and other probing techniques has revealed a more nuanced picture of syntactic representation, showing that although LLMs do encode syntactic information, they are also sensitive to the interference of local cues like closely occurring words (Agarwal et al., 2025; Simon et al., 2025). However, models can still pick up structural differences in sentences that look similar on the surface, and capture the subtle meaning changes they produce, showing an ability to integrate syntactic and semantic processing (Kennedy, 2025). Minimal-pair probing provides a useful way to track specific aspects of linguistic processing and representations in minimally varying contexts: He et al. (2024) compared how models process minimal pairs contrasting a well-formed sentence with an otherwise identical sentence including a targeted grammatical violation. Through this method, they showed that some information about syntactic competence is represented in early layers, and that as sentences become more complex, the models need more layers to evaluate their grammaticality. Crucially, features at the interface between syntax and semantics are more difficult for models to learn than purely syntactic patterns (Graichen et al., 2026).

Linguistic complexity Linguistic complexity can be examined from two complementary perspectives. On the one hand, it can be understood as the degree of elaboration of a grammatical system at the different levels of representation, often known as formal complexity. On the other hand, it can be defined in terms of the effort required by language users to process or learn a given linguistic structure, often known as functional or processing complexity. Formal complexity pertains to the structural and computational properties of grammar. This involves, for instance, the number of rules, the degree of hierarchical embedding, the number of idiosyncrasies in the system, or the length and depth of constituent and sentence structure (Culicover, 2014; Hawkins, 2014; Trotzke and Zwart, 2014). Functional or processing complexity, in turn, concerns how such structures are implemented and experienced by language users, shaping parsing, memory, acquisition, and neural activity (Hawkins, 2014; Menn and Duffield, 2014). The two perspectives are interrelated: formal descriptions delimit what structures a grammar makes available, while functional accounts reveal how these structures are deployed and constrained in real-time language processing. Given the strong interplay between these dimensions of complexity, we will treat complexity holistically, assuming that a given structure is more complex than its baseline irrespective of the specific type of complexity involved. We nonetheless return to the distinction between formal and functional complexity in the Discussion.

Contrast	Example
Coordination	(1) The blacksmith is babbling and the politicians are doubting and the tutor is writing and the banker is listening
Subordination	(2) The blacksmith is babbling that the politicians are doubting that the tutor is writing that the banker is listening
Right branching	(3) The politicians advised the potters that were waiting
Center embedding	(4) The potters that the politicians advised were waiting
Unambiguous	(5) The boy of the mother who gave birth got cold.
Ambiguous	(6) The friend of the mother who gave birth got cold.

Table 1: Examples of the experimental manipulations in the three main datasets.

3 Experimental materials

We designed three main datasets instantiating the complexity contrasts shown in Table 1.[1] These contrasts are implemented through classic minimal pairs, allowing us to isolate the contribution of each linguistic manipulation. We report results for these datasets in the main text, and present additional dataset variants and analyses in the appendices.

Coordination vs. subordination These clause-combining operations yield syntactic configurations differing in hierarchical depth. Coordinated sentences are the result of an iterative process generating a flat structure. Subordinated sentences result from a recursive process generating an embedded structure (examples (1) and (2) in Table 1). In this sense, subordination corresponds to a higher level of *formal* complexity (Trotzke and Zwart, 2014). In spite of the near-total lexical overlap, the hierarchical differences between coordinated and subordinated structures lead to clear meaning differences. For example, since coordination links clauses of equal syntactic status, swapping them leaves meaning essentially unchanged. In contrast, the hierarchical dependency created by embedding one clause within another becomes evident when the clauses are swapped. Our primary datasets consist of sentences which only vary in the conjunction used (*and* vs. *that*). Details are in App. A.1. Supplementary manipulations (App. C.4) involve varying the number of clauses that are combined (from 2 to 4) and contrasting *or* (instead of *and*) with *that*.

Right-branching vs. center-embedding These structures (examples (3) and (4) in Table 1) differ sharply in their processing-related, or *functional*, complexity, despite being formally similar in terms of hierarchical depth and the type of underlying dependency relations: the two contain a relative clause (RC) modifying one of the NPs in the main clause. Yet, while right-branching sentences extend linearly by placing the RC at the end, the RC in center-embedding sentences appears between the subject and the verb, creating a long-distance dependency. Even though this alternation leaves sentence meaning essentially intact (differing only in information-structural choices about which noun phrase is in focus), a vast amount of psycholinguistic literature has shown that center-embedding sentences are harder to process due to increased integration and memory storage cost (e.g., Gibson, 1998; Lewis and Vasishth, 2005). Our dataset creation procedure is detailed in App. A.2. A further experiment (App. C.5) instantiates a similar complexity contrast using subject vs. object relative clauses, with the former representing the less complex condition.

Unambiguous vs. ambiguous RC attachment is a common source of ambiguity that arises when a RC can modify more than one preceding NP. This ambiguity can be eliminated when only one NP is semantically compatible with the content of the RC, illustrating how cues at the syntax–semantics interface guide attachment decisions. Our datasets contain minimal pairs differing only in whether the attachment is lexically disambiguated in favor of the closer NP (example (5) in Table 1, a case of so-called "low" attachment) or whether both NPs remain semantically plausible, thereby sustaining the ambiguity (example (6) in Table 1).[2] Although the two sentences are formally parallel, many psy-

[1] Our datasets are available from: https://github.com/franfranz/syntactic_complexity_in_LLMs

[2] We also include an unambiguous condition favoring attachment to the first NP, so-called "high" attachment (App. A.3). We focus on the comparison between ambiguous and low-attachment sentences because low attachment reflects the typical attachment preference in English, and thus provides a natural baseline. High-attachment sentences behave similarly to low-attachment ones (App. C.6).

cholinguistic studies have shown that semantically unambiguous RCs permit faster attachment, and are thus less complex, whereas ambiguous ones generate processing slow-downs due to increased competition between attachment sites (e.g., Gibson et al., 1996; Carreiras and Clifton, 1999). We thus consider this complexity contrast more *functional* in nature. Dataset details are in App. A.3.

Higher complexity elicits higher LLM surprisal We ran a preliminary check that LLM processing aligned with the predictions from the (psycho)linguistic literature sketched above. Indeed, for all datasets, all LLMs showed a higher per-token surprisal for the more complex condition as determined by a one-sided t-test ($\alpha = 0.05$). Exact surprisal values are given in Table 5 of App. C.1.

4 Methods

Given an LLM and a complexity contrast (e.g., coordination vs. subordination), we examine whether and how the evolution of representation geometry over layers differs between the hard" and easy" conditions. To track this evolution, we focus on the residual stream, the main representational pathway that is progressively updated through residual connections. To do so, for each dataset-model pair, we first extract the layerwise last-token embeddings in the residual stream (Elhage et al., 2021). We use last-token representations as they are the only ones to attend to the entire sequence, and the ones used to predict the next token. For a dataset of size N and an LLM with hidden dimension D, this yields a sequence of representations in $\mathbb{R}^{N\times D}$. We compute the ID and between-contrast representational similarities using these representations.

Intrinsic dimension In order to compare LLM processing complexity profiles in different conditions, for each model-dataset-layer combination, we compute the ID of the representations. Among the many ID estimators developed (Campadelli et al., 2015), we choose the TwoNN estimator (Facco et al., 2017), as it makes minimal assumptions (only local data uniformity up to the second nearest neighbor), correlates highly to other estimators (Cheng et al., 2023), and has been widely used in past work estimating the ID of data representations (Valeriani et al., 2023; Chen et al., 2024; Lee et al., 2025). Details are in App. B.2.

Representational similarity Because we consider data *manifolds*, we are interested in a notion of *local* representational similarity. We make use of the Information Imbalance (Δ) (Glielmo et al., 2022), a measure of neighborhood divergence between representation spaces. Δ is *asymmetric*: in general, $\Delta(A \to B) \neq \Delta(B \to A)$. As such, Δ can be thought of as a *directional* distance between spaces. Intuitively, it quantifies the amount of information about one space (e.g., the space of coordinated sentences) that is also captured in another space (e.g., the space of matched subordinated sentence). Both Cheng et al. (2025) and Acevedo et al. (2026) found Δ to provide a clearer signal than the more widely used Centered Kernel Alignment measure (Kornblith et al., 2019) in the context of comparing LLM representations. Details are in App. B.3.

Models We test six LLMs in sizes ranging from 7 to 14 billion parameters: Gemma-2-9b (Gemma) (Riviere et al., 2024), Llama-3-8B (Llama) (Grattafiori et al., 2024), OLMo-2-13B (OLMo) (Walsh et al., 2025), Mistral-7B-v0.1 (Mistral) (Jiang et al., 2023), Pythia-12B (Pythia) (Biderman et al., 2023), and Qwen-2.5-14B (Qwen) (Yang et al., 2025). These models may differ in specific architectural choices, for instance nonlinear activation or positional embedding, but all consist of decoder layers that each include an attention and feedforward module. Our use of residual stream representations abstracts away from these specific architectural choices to allow high-level comparison of the models' layerwise dynamics.

5 Experiments

Our observations are always based on inspecting the behavior of all six studied LLMs. However, for space reasons, we display the results for Llama, OLMo and Pythia in the main text, and Gemma, Mistral and Qwen in appendices.

5.1 Intrinsic dimension

Generic sequences We build on Cheng et al. (2025), who showed that, when fed random naturalistic input from a corpus, LLMs systematically display a profile with (at least) one ID "peak" that marks an area of deep linguistic processing. We replicate this result in App. C.2, extending it to 3 LLMs that were not studied by Cheng and colleagues. In the same appendix, we confirm their observation that there is a broad alignment between the ID peaks and top performance phases in three semantic and syntactic probing tasks. In all the

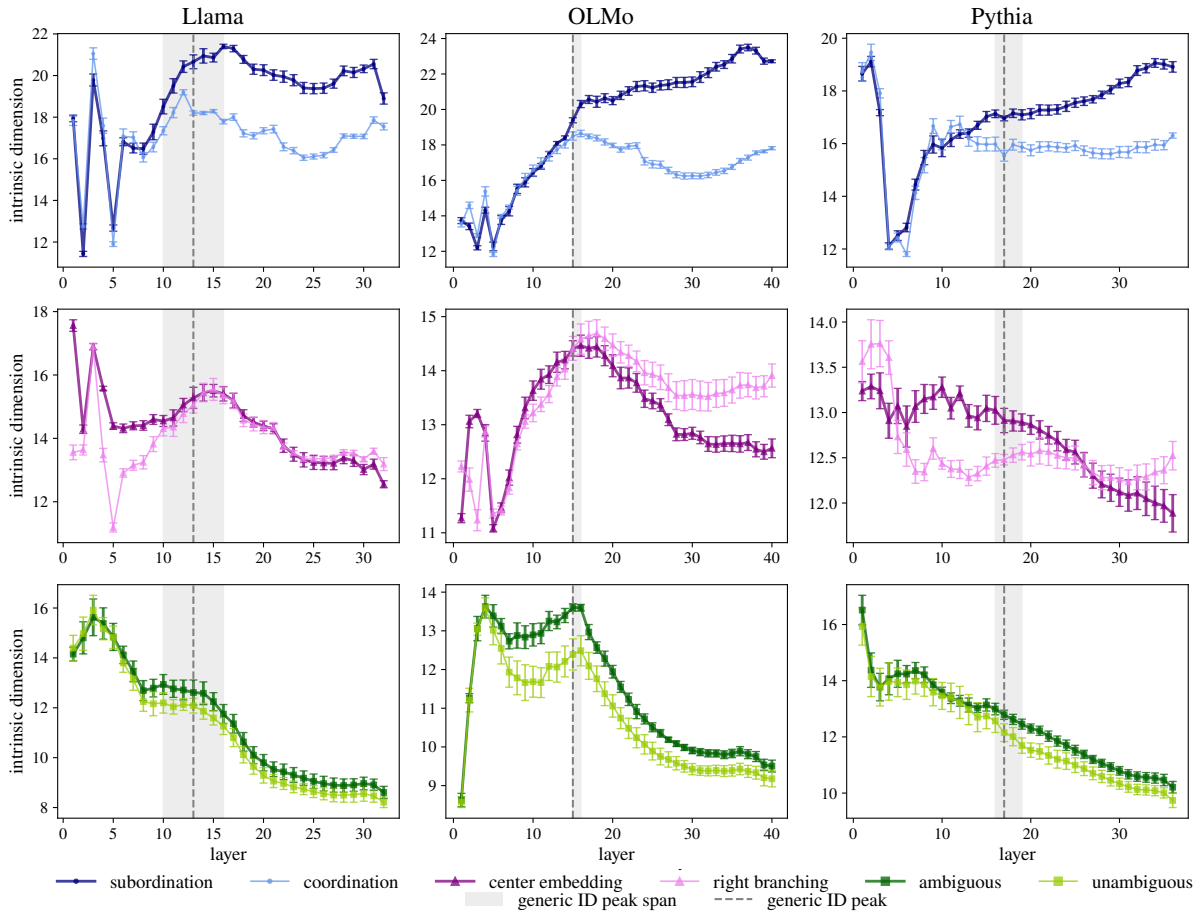


Figure 1: ID profiles through LLM layers (means and error bars across 5 partitions). Vertical dashed line marks maximum ID on generic sequences, and shaded area the corresponding span, estimated as explained in App. C.2.

following experiments, we thus take the generic-sequence ID-peak span across LLM layers (automatically identified as explained in App. C.2) as a reference point for a phase in which the model is focusing on deeper linguistic processing of the inputs, as opposed to surface-level input processing or output generation.

Before analyzing our contrast-specific datasets, we note that, in absolute terms, the maximum ID of the various datasets is always orders of magnitudes smaller than the ambient dimension, and it largely correlates with the length of the inputs. It is highest for the 100-word-long generic sequences, next-higher for the 18-word-long 4-clause coordinated/subordinated sentences, and then lower and comparable for the right-branching/center-embedding inputs (8 words) and the unambiguous/ambiguous inputs (13 words on average).[3]

Coordination vs. subordination The first rows of Fig. 1 and Fig. 6 (App. C.3) show how ID evolves for this contrast across layers.[4] After some initial fluctuations possibly due to estimation noise, we observe a very consistent pattern across LLMs: the coordination and subordination ID curves first overlap, but they eventually diverge, typically under or just before the generic-ID peak (considerably earlier for Qwen only). From that point onward, ID is clearly higher for the more complex subordination condition than for coordination.

Right-branching vs. center-embedding This contrast is shown in the middle rows of Fig. 1 and Fig. 6 (App. C.3). After some initial fluctuation, we observe the more complex center-embedding condition to reach a higher ID than the right-branching one. However, this distinction is present only until the curves reach the generic-ID peak (for Pythia,

[3]The ID of the the ambiguity pairs is comparable to that of the slightly shorter right-branching/center-embedding pairs, if not lower, probably because the ambiguity datasets contain less data-points than the others (App. A.3). In any case, our analysis focuses on the relative ID *profile* over layers, rather than on absolute ID values.

[4]Figures 7 and 8 (App. C.4) show ID profiles for different numbers of clauses being combined, and when clauses are coordinated with *or* instead of *and*, respectively.

a bit later). After the generic-ID peak, the two curves merge or, for 4/6 models, there is even a *reversal*, whereby right-branching ID is higher. Intriguingly, right-branching sentences end with a nested clause ("The politicians advised the potters [that were waiting]"), unlike the center-embedding ones ("The potters [that the politicians advised] were waiting"). This suggests that the higher right-branching ID after the generic-ID peak might be related to the presence of a nested clause at the end of the sentence, analogously to what we just saw for subordination (recall that we are measuring the ID of the *last* token, that will be inside the nested clause in right-branching sentences only).

Unambiguous vs. ambiguous Results are in the last rows of Fig. 1 and Fig. 6. We observe a general downward trend of ID, that in most cases becomes clearer after the generic-ID peak. We do not attribute this pattern directly to the attachment manipulation. Unlike in the previous datasets (coordination vs. subordination and right-branching vs. center-embedding), where sentence structure is tightly controlled and the final token is always a present participle, the sentences in the present datasets display more variation in the final tokens (e.g., they may be nouns, adverbs, or adjectives: see Table 4, App. A.3). Although such factors are controlled within each minimal pair, allowing us to interpret differences in ID profiles between conditions, they vary across pairs. As a result, they may shape the overall ID trajectory independently of the contrast under investigation. Notably, the consistently higher ID observed for the ambiguous condition suggests that the models are sensitive to the presence of ambiguity, thus supporting the use of this dataset to probe how LLMs represent it.

More directly relevant to our investigation, we observe a general tendency for the ID of the more complex ambiguous condition to dominate the simpler unambiguous condition. In this case, however, the generic-ID-peak phase is not acting as a clear delimiter between different ID patterns, and the difference between the curves tends to be small.

In sum, ID confirms our predictions based on the (psycho)linguistic literature, but in very distinct ways for the three contrasts considered. Syntactic nesting emerges as the linguistic phenomenon most consistently affecting ID profiles: processing deeply nested structure plausibly requires more complex features, leading to higher ID profiles for subordinated structures. Further, the differentiation between flatter and more nested sentences emerges over the layers that coincide with the generic-ID peak, which was proposed as a signature of deeper linguistic processing. It figures that nested structure processing is part of this phase.

ID also responds to long-distance agreement resolution, but in this case the complexity effect is visible earlier, typically before the generic-ID peak. From a strictly structural point of view, center-embedded and right-branching sentences are identical, and their meaning only differs in terms of which element is foregrounded. Thus, the early ID differentiation might be a marker of a lower-level, word-bound kind of processing, linked to matching agreement features between the main subject and its predicate. Intriguingly, for most models we observe higher post-generic-peak ID for right-branching sentences, which we linked to their final nesting, thus relating it to the ID increase of the subordinated sentences above.

Finally, yet another ID pattern appears for ambiguity resolution. Ambiguous and unambiguous profiles are similar, but the ambiguous constructions have higher ID across the board. This pattern is open to multiple interpretations, but it suggests a contrast that is not linked to specific phases of linguistic processing, but rather to the instability triggered by ambiguity (e.g., the higher complexity could be triggered by keeping both sentence interpretations open until the end).

5.2 Representation similarity across layers

Our claim that ID profiles reflect a genuine difference in LLM processing of different complexity types is supported by a follow-up experiment in which we tracked representational similarity between matched subordinated vs. coordinated and right-branching vs. center-embedding sentences.[5]

The top rows of Figures 2 (and 13 in App. C.8) show the evolution of the Information Imbalance Δ across layers for the coordinated and subordinated sentence datasets. Δ is generally extremely low in both directions, i.e., not surprisingly, the model representations of sentence pairs such as those in Table 2 (App. A.1), that share word order and virtually all lexical materials, are extremely

[5]The sentences in the ambiguity datasets have significant differences in lexical material, and thus they cannot be naturally paired to measure representational similarity. In App. C.7, we provide independent empirical validation for the Information Imbalance measure we use here, showing how it approaches 0 for near paraphrases, and 1 for unrelated sentences.

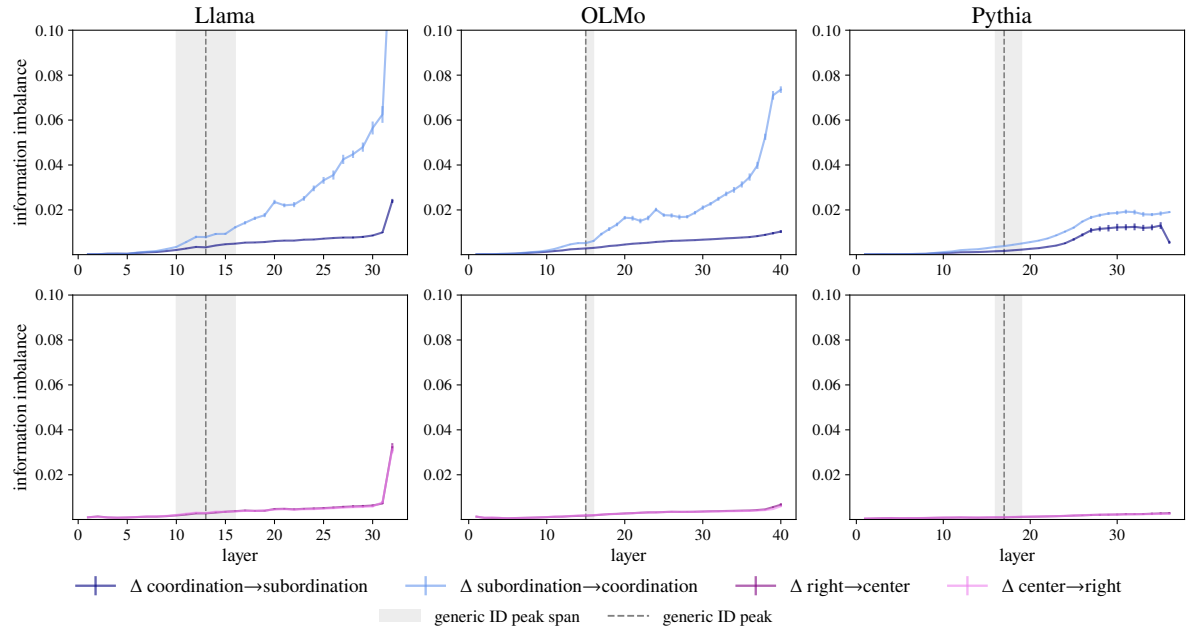


Figure 2: Information Imbalance Δ between coordinated/subordinated (top) and right-branching/center-embedding sentences (bottom): means across 5 partitions with error bars (present, but hardly visible). Shaded area marks generic ID-peak span, with a vertical dashed line at the generic-ID maximum. Higher Δ means lower similarity.

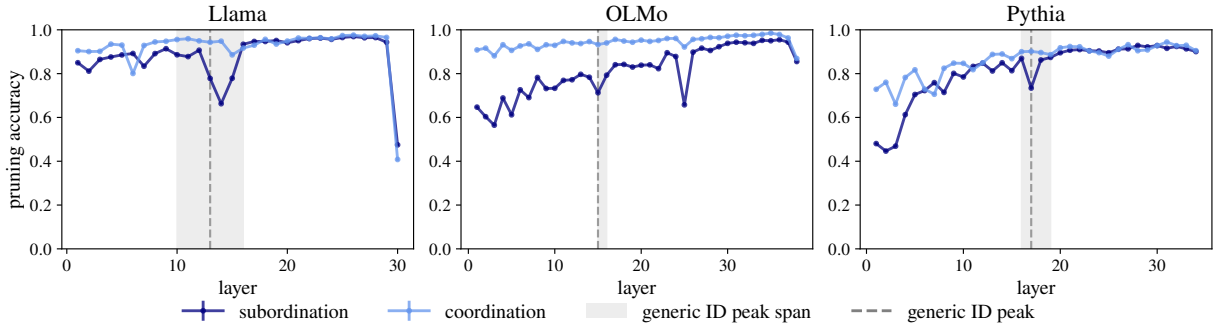


Figure 3: Pruning accuracy (proportion of cases in which pruning a layer did not change next-token prediction) of **coordinated**/**subordinated** sentences. Means and (hardly visible) standard error bars over 5 partitions.

similar. To put this degree of similarity into perspective, according to the theoretical simulations of Acevedo et al. (2026), a Δ of 0.1 (the largest value on our y axes) corresponds to 90% of shared features. There is also, however, a clear asymmetric divergence in Δ profiles for 5/6 models. Just under the generic-ID peak, and thus just where the ID difference between the coordination and subordination sets emerges, Δ(*coord*→ *subord*) becomes progressively *larger* than Δ(*subord*→ *coord*), except for Qwen (Fig. 13), where the generic-ID peak only marks a phase in which representations diverge, but there is no asymmetry. Thus, in general, the representations of the two sentence types grow apart, and the information contained in the coordinated sentences predict increasingly *less* that of the corresponding subordinated sentences. This makes intuitive sense: if I say that "*The blacksmith is babbling that the politicians are doubting*", I am approximately providing the same information as in the corresponding coordinated sentence ("*The blacksmith is babbling and the politicians are doubting*"), but on top of that I am also specifying *what* it is that the blacksmith is babbling, a piece of information that is not contained in the coordinated sentence.

Looking next at the bottom rows of the figures (center-embedding vs. right-branching), we again observe very strong similarity, decreasing only slightly across the layers. Importantly, in this case there is no asymmetry. This suggests that the LLMs capture the fact that pairs such as "*The potters that the politicians advised were waiting*" and "*The politicians advised the potters that were*

waiting" carry the same denotational meaning.

Finally, in both cases, we observe that Δ sometimes spikes on the very last layers. This is possibly due to the fact that, as the models must switch to predicting a concrete next token on these layers, and this is likely different for the compared conditions, they must represent different superficial information at the very end.

In sum, at the generic-ID-peak, the LLMs start building structures that make subordinated and coordinated sentences diverge in representation, with the representations of the more richly structured subordinated sentences becoming more predictive of the simpler coordinated sentences than the reverse. On the other hand, no asymmetry is observed for the center-embedding vs. right-branching contrast, and only a very weak divergence in Δ across the layers, in line with the strong semantic and structural similarity between the conditions. Again, this suggests a stronger sensitivity of the LLMs to the nesting contrast in their deeper layers.

5.3 Pruning experiments

We perform an intervention experiment, looking at the effect of pruning layers that the ID and Δ tests singled out as crucial for complexity processing. In practice, we remove in turn each layer of an LLM, so that the output from the previous layer is directly fed into the next one. As standard in prior work (Gromov et al., 2025; Lad et al., 2025; Csordás et al., 2025), we iteratively prune each layer l by removing layer l's attention + MLP contribution, and directly feeding residual stream representations from the previous layer $l-1$ to the next layer $l+1$. We then compute a *pruning accuracy* score by counting the proportion of inputs for which the pruned LLM predicts the same next token as the intact network. Note that, when the pruned-model prediction is different from the original one, there is no guarantee that the latter is better. However, the measure still captures the extent to which pruning changes an LLM behaviour.[6]

Fig. 3 (and Fig. 14 in App. C.9) show pruning profiles for the coordination/subordination conditions. For all models, we observe a tendency for pruning to have a stronger effect on subordinated than coordinated sentences, in line with the other markers of extra complexity for this condition (the effect is typically clearer in the earlier layers). Strikingly, for 4/6 LLMs there is a clear dip in accuracy when pruning layers in the the generic-ID peak phase, bringing further support for the hypothesis that this is a stage in which different LLMs are performing structure-building operations that are especially important for highly nested sentences. Of the two outliers (Fig. 14 in the appendix), Qwen shows a subordination-specific dip just after the generic-ID peak, whereas in the case of Gemma there is a dip late in the peak, but it is actually stronger for coordination.

Moving to the other contrasts, Figures 15 and 16 (App. C.9) do not show differential pruning effects across conditions (right-branching vs. center-embedding and unambiguous vs. ambiguous), and there are no layer-specific dips, except for strong initial- and final-layer effects.

To sum up, we confirm the special status of nesting-driven complexity, and the crucial role that the layers under the generic-ID peak have in determining nested input processing.

6 Discussion

We showed that different pre-trained LLMs display similar ID profiles, and that the latter are signatures of consistent processing stages. We refined this observation by showing that generic-ID peaks, previously conjectured to be the locus of abstract linguistic processing, correspond to areas where the contrast between flatter (coordinated) and more nested (subordinated) structures emerge. The other contrasts we studied (right branching vs. center embedding and unambiguous vs. ambiguous relative clause attachment) also have consistent ID signatures, that however differ and are less clearly related to the generic-ID peak stage. Intriguingly, as discussed in Sec. 3, the distinction between subordination and the other complexity factors corresponds to a distinction between a more formal type and two more functional ones. Future work should conduct more case studies along this divide, to ascertain whether the ID differences we found are general properties of formal vs. functional complexity.

From an LLM interpretability perspective, our results shed light on the mechanisms behind the consistent ID profiles that are observed across model layers, and contribute to recent debates on the "universality" of LLM representations (Huh et al., 2024), suggesting that the conjectured universality is at least in part due to similarities in the ways in

[6] In App. C.10, we report similar results obtained with a continuous pruning score, namely KL divergence.

which LLMs organize linguistic processing. From a linguistic point of view, we have shown that ID estimation in LLM representations can provide a new tool to track different degrees of linguistic complexity, a task that is still seen as an open problem in this area (Newmeyer and Preston, 2014).

We believe that, even if LLMs cannot be straightforwardly treated as models of human cognition (Ziv et al., 2026), they still can serve as probes for understanding the structure and complexity of linguistic signals. From this perspective, ID profiles can reveal properties of language representation that are relevant to cognitive science, even if the exact mechanisms giving rise to the representations might be different from those found in humans.

Limitations

Due to compute restrictions, we cannot extend our experiments to larger language models than the ones we studied. However, future work should ascertain the extent to which the observed patterns depend on model size, for families of related LLMs, and track their emergence during training, for LLMs that make their intermediate checkpoints available.

Our results should also be reproduced in languages other than English, and with more naturalistic data than the one we used here. Even in English, we have only started scraping the surface of the set of possible linguistic phenomena that could be studied under the ID lens, and future work should expand the list of case studies to demonstrate that the relation between increases in ID and linguistic complexity is truly general.

Finally, while the pruning experiments provide a proof of concept that there is a causal link between the presence of an ID peak on a layer and its impact on model behavior, the most important aim of future work should be to establish a clear mechanistic explanation of the relation between intrinsic dimension of representations and the linguistic complexity.

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A Datasets

A.1 Coordination vs. subordination

Using OpenAI's GPT-4 followed by manual editing, we first built: i) a list of 17 propositional verbs that could also be used as intransitives, such as *babbling* and *dreaming*; ii) a list of 65 pure intransitive verbs taking human subjects (such as *shivering* and *sleeping*); and iii) a list of 74 nouns, divided between 30 proper nouns (*Michael, Sarah*) and 44 profession names (*gardener, programmer*). We randomly sampled elements from these lists to generate 50k 4-clause subordinated and coordinated sentence pairs that matched the following template:

```
NP1 PROPVERB1 CONJ NP2 PROPVERB2 CONJ
NP3 PROPVERB3 CONJ NP4 INTVERB
```

where the NPs are either proper names or noun phrases formed by *the* followed by a profession noun (randomly in singular or plural form); the PROPVERBs are propositional verbs in the present continuous (in singular or plural form in agreement with the subject); the INTVERB is a pure-intransitive verb, also in the present continuous and agreeing with its subject; and CONJ is either *that* or *and/or*. The words are sampled so that no noun or verb lemma is repeated. In order to maintain uniqueness when removing the clauses in the middle (see below), we also ensured that the tuples formed by <NP1, PROPVERB1, NP4, INTVERB> and <NP1, PROPVERB1, NP2, PROPVERB2> were unique. Five random examples from the dataset constructed in this way are shown in Table 2

In order to build the shorter 3- and 2-clause sentences analyzed in App. C.4, we simply removed one or two clauses, respectively, from the middle of the 4-clause sentences. We split the datasests into partitions of 10k sentences, and repeat all experiments 5 times, always reporting means and standard errors across the partitions.

A.2 Right-branching vs. center-embedding

We used the same nouns and intransitive verbs as for the coordination/subordination datasets (see App. A.1). By again querying ChatGPT 4 and manually editing its outputs, we created a list of 100 transitive verbs that take human subjects and objects (*betray*, *fascinate*, *scold*...). We constructed a set of 50K matched center-embedding and right-branching sentences by randomly sampling from these lists according to the following templates:

```
CENTER EMBEDDING:
NP1 that NP2 TRVERB INTVERB
RIGHT BRANCHING:
NP2 TRVERB NP1 that INTVERB
```

where NP1 is formed by *the* followed by a profession noun in plural or singular form; NP2 is either formed in the same way or it is a proper noun; TRVERB is a transitive verb in past-simple form; and INTVERB is a past continuous form of an intransitive verb agreeing in number with NP1. Again, no sentence contains a repeated noun or verb. Using the same data, we constructed an additional contrast between subject relative clauses (SRC) and object relative clauses (ORC). This contrast was derived directly from the center-embedding items. The ORC condition corresponds exactly to the center-embedding dataset, relabeled here to make the contrast in terms of relativization type explicit, rather than the branching configuration. The SRC sentences were created by promoting the relative clause subject to subject position in the main clause, while preserving the predicate and lexical content. Five random examples for each of the two contrasts are given in Table 3.

We split the datasests into partitions of 10k sentences, and repeat all experiments 5 times, always reporting means and standard errors across the partitions.

A.3 Unambiguous vs. ambiguous

This dataset contains three attachment conditions: an ambiguous condition, in which both noun phrases are equally plausible relative-clause (RC) antecedents, and two unambiguous conditions, in which semantic biases strongly favor attachment

Quinn is rejoicing and/or/that the surgeon is doubting and/or/that Mary is screaming and/or/that the driver is faltering
The doctors are muttering and/or/that the firefighter is babbling and/or/that Bill is complaining and/or/that the consultants are hesitating
The engineers are singing and/or/that Jordan is dreaming and/or/that the tutors are rejoicing and/or/that the soldier is sliding
The artist is remembering and/or/that Matthew is doubting and/or/that the judge is writing and/or/that Taylor is trembling
Emily is complaining and/or/that Casey is mumbling and/or/that the manager is writing and/or/that the blacksmith is shivering

Table 2: Examples from the 4-clause coordination/**subordination** datasets. The 3-clause sentence derived from the first example in this table is: "Quinn is rejoicing and/that the surgeon is doubting and/that the driver is faltering". The 2-clause sentence from the same example is: "Quinn is rejoicing and/that the driver is faltering".

Right-branching	*Center-embedding*
Sarah intimidated the potters that were frowning	The potters that Sarah intimidated were frowning
James harassed the veterinarians that were sulking	The veterinarians that James harassed were sulking
Bill excluded the driver that was escaping	The driver that Bill excluded was escaping
Elizabeth praised the foresters that were chuckling	The foresters that Elizabeth praised were chuckling
The gardeners applauded the blacksmith that was hurrying	The blacksmith that the gardeners applauded was hurrying
Subject relative clause	*Object relative clause*
The politicians that were waiting advised the potters	The potters that the politicians advised were waiting
The programmer that was gasping rejected the pharmacist	The pharmacist that the programmer rejected was gasping
The blacksmith that were relaxing punished the doctors	The doctors that the blacksmith punished were relaxing
The astronomer that were pacing appointed the blacksmiths	The blacksmiths that the astronomer appointed were pacing
The mechanics that was persisting invited the politician	The politician that the mechanics invited was persisting

Table 3: Examples from the right-branching/**center-embedding** and the subject relative clauses/**object relative clauses** datasets.

to either NP1 (high attachment) or NP2 (low attachment). All sentences conform to the following template:

`NP1 of NP2 who RC CONTINUATION`

In the main text, we focus on the contrast between ambiguous and low-attachment sentences because English favors low attachment, and, thus, this condition displays the expected and least demanding interpretation, making it a natural baseline. Results including the high-attachment dataset are in App C.6.

Using ChatGPT 4, we generated a list of NPs, relative clauses, and continuations that would encode four types of semantic bias: age, gender, role, and logical contradiction. Bias-specific elements were generated using controlled prompting and subsequently filtered, ranked, and manually validated to ensure semantic clarity, grammaticality, and compatibility across conditions. To construct the final dataset, validated elements were recombined under strict constraints that prevent excessive lexical overlap across sentences, while preserving bias consistency. This procedure generated a total of 10,880 attachment triplets (32,640 sentences), where each triplet shares the same relative clause and continuation. Examples for gender- and age-based semantic disambiguations are given in Table 4 and further details can be found in Mathewos and de Dios-Flores (2026).

The sentences were split into 5 equal partitions of 2176 items each. We repeat all experiments 5 times, always reporting means and standard errors across the partitions.

B Methods

B.1 Code and compute estimates

Code To compute intrinsic dimensionality and information imbalance, we used the DadaPy toolkit. Representation extraction was accomplished by trivially adapting the code made available by Cheng et al. (2025). Surprisal was computed using the Python `surprisal` package. In all cases, code was run with default parameters.

URLs and licenses of the used assets are provided in the following list:

Gemma https://huggingface.co/google/gemma-2-9b; license: gemma

Llama https://huggingface.co/meta-llama/Meta-Llama-3-8B; license: llama3

Mistral https://huggingface.co/mistralai/Mistral-7B-v0.1; license: apache-2.0

OLMo https://huggingface.co/allenai/OLMo-2-1124-13Bhttps; license: apache-2.0

Pythia https://huggingface.co/EleutherAI/pythia-12b-deduped; license: apache-2.0

Qwen https://huggingface.co/Qwen/Qwen2.5-14B; apache-2.0

Cheng et al.'s code https://github.com/chengemily1/id-llm-abstraction; license: MIT

DadaPy https://github.com/sissa-data-science/DADApy; license: apache-2.0

Probing tasks https://github.com/facebookresearch/SentEval/tree/main/data/probing; license: bsd

surprisal https://github.com/aalok-sathe/surprisal; license: MIT

Compute Representation extraction took a few wall-clock hours for each LLM, on 1 or 2 NVIDIA A30 GPUs. The most time-consuming extraction step was to run the pruning experiments, as different sets of representations had to be extracted after removing one layer at a time from each of the models, resulting in about two days of running per model. Probing experiments took between 1 and 2 days per task, running on CPUs. Compute time for the other experiments was negligible.

B.2 TwoNN ID estimation

The TwoNN estimator belongs to a class of so-called "geometric" ID estimation methods (Campadelli et al., 2015). This class, which also includes, a.o., the Maximum Likelihood Estimator of Levina and Bickel (2004), to which TwoNN highly correlates in practice (Cheng et al., 2023), generally works as follows. First, under distributional or regularity assumptions of the space, properties of points on the manifold behave according to some theoretical distribution. The analytic expression of this distribution depends on the ID. Then, the ID can be estimated via maximum likelihood estimation from data.

Age bias	*Gender bias*
Ambiguous: The neighbor of the grandpa who paid a mortgage stood nearby.	**Ambiguous**: The sister of the heiress who was menstruating cooked rice.
Low attachment: The child of the comrade who paid a mortgage stood nearby.	**Low attachment**: The uncle of the maiden who was menstruating cooked rice.
High attachment: The uncle of the child who paid a mortgage stood nearby.	**High attachment**: The maiden of the uncle who was menstruating cooked rice.

Table 4: Examples from relative-clause unambiguous/**ambiguous** attachment datasets. Low attachment is the unambiguous condition used in the main-text experiments.

TwoNN realizes these steps as follows. Assume points on the underlying manifold are distributed as a locally homogeneous Poisson point process, where "locally" means up to the second nearest neighbor of each point. Let $\delta_k^{(i)}$ be the Euclidean distance between the point x_i and its kth nearest neighbor. Then, the distance ratios $\mu_i := \delta_2^{(i)}/\delta_2^{(i)} \in [1, \infty)$ have the cumulative distribution function $F(\mu) = (1-\mu)^d \mathbf{1}[\mu \geq 1]$ (Facco et al., 2017). The ID, given by d, is estimated as $d = -\log(1 - F(\mu))/\log \mu$ via maximum likelihood estimation over all data points.

B.3 Information Imbalance

Consider two representation spaces A and B consisting respectively of paired data points $\{x_i\}_{i=1}^N$ and $\{y_i\}_{i=1}^N$. Let r_{ij}^X refer to the neighbor-rank of point j to point i in space X. For instance, if x_j is x_i's first nearest neighbor in space X, then $r_{ij}^X = 1$. Then, $\Delta(A \to B)$ is given by (Glielmo et al., 2022):

$$\Delta(A \to B) := \frac{2}{N^2} \sum_{i=1}^{N} \sum_{j=1}^{N} r_{ij}^B \mathbf{1}[r_{ij}^A = 1]. \quad (1)$$

If nearest neighbors in A are also nearest neighbors in B, then $\Delta(A \to B) \approx 0$. If nearest neighbors in A have a uniformly-distributed neighbor rank in B, i.e., neighbors in A are uninformative about neighbors in B, and $\Delta(A \to B)$ is near 1.

C Additional results

C.1 Surprisal

For each dataset, we sampled 1,000 sentences at random to estimate the mean per-token surprisal under each LLM. We compute the mean surprisal per sentence, then average over all 1,000 sentences to obtain one surprisal value per-sentence. For each model-dataset combination, the distribution of surprisals over the 1,000 sentences was approximately normal, according to a Shapiro-Wilk test with a conservative p-value cutoff of $\alpha = 0.1$. We then performed a one-sided difference-of-means t-test between the **more** and less complex conditions, finding in all cases that the LLM has higher surprisal on the **more** complex condition, significant at $\alpha = 0.05$. Results are displayed in Table 5.

C.2 Intrinsic dimension of generic sequences

ID profiles We reproduce the finding by Valeriani et al. (2023) and Cheng et al. (2025) that, on generic in-distribution data, a peak in the intrinsic dimension of LLM activations emerges in the intermediate layers. To do so, we use a sample of 50k sequences from the Wikitext corpus (Merity et al., 2017) repurposed from Macocco et al. (2025). Each sequence consists of 100 words and begins with a start of sentence but is not constrained to end with a period, so that the final word can have any part of speech. Then, for each model, on five random non-overlapping data splits of 10k sequences each, we compute the ID using the TwoNN estimator on the last-token representation. This produces an ID profile across layers for each model, shown in Fig. 4. Finally, for each model, following Cheng et al. (2025), we heuristically demarcate the first ID peak-span (gray in the figures). We locate the nearest inflection points around the maximum using second-order finite differences. Note that, with respect to Cheng and colleagues, we extend these experiments to Gemma, Qwen and a newer and larger version of OLMo, thus further confirming the generality of their observation.

Fig. 4 first of all confirms that the intrinsic dimension is always orders of magnitude smaller than the ambient ones (which is always $> 3.5K$). Importantly, all models have an intrinsic dimensionality peak in their mid layers, followed in some cases (OLMo, Pythia, Qwen) by another late-layer peak, a phenomenon also reported by Cheng and colleagues.

Model	subordination/coordination		center embed/right branch		ambiguity		
	coord.	**subord.**	right	**center**	low	high	**ambiguous**
Llama	$5.39_{0.01}$	$\underline{5.65}_{0.01}$	$7.74_{0.02}$	$\underline{7.98}_{0.02}$	$6.07_{0.02}$	$6.08_{0.02}$	$\underline{6.11}_{0.02}$
OLMo	$4.87_{0.01}$	$\underline{5.11}_{0.02}$	$6.51_{0.02}$	$\underline{6.75}_{0.02}$	$5.10_{0.02}$	$5.16_{0.02}$	$\underline{5.18}_{0.02}$
Pythia	$4.77_{0.01}$	$\underline{5.14}_{0.01}$	$6.66_{0.02}$	$\underline{7.06}_{0.02}$	$5.37_{0.02}$	$5.39_{0.02}$	$\underline{5.39}_{0.02}$
Gemma	$6.86_{0.02}$	$\underline{7.62}_{0.02}$	$10.35_{0.04}$	$\underline{10.58}_{0.04}$	$6.79_{0.03}$	$6.75_{0.03}$	$\underline{6.90}_{0.03}$
Mistral	$4.87_{0.01}$	$\underline{5.22}_{0.02}$	$6.86_{0.02}$	$\underline{7.00}_{0.02}$	$5.50_{0.02}$	$5.52_{0.02}$	$\underline{5.58}_{0.02}$
Qwen	$4.77_{0.01}$	$\underline{5.05}_{0.01}$	$6.44_{0.02}$	$\underline{6.61}_{0.02}$	$5.13_{0.02}$	$5.15_{0.02}$	$\underline{5.17}_{0.02}$

Table 5: Average surprisal of contrast datasets. The mean surprisal per-token (nats) ± 1 SE is shown for the six LLM rows, for each linguistic phenomenon column and contrast condition subcolumns. Each value is computed from a 1,000-sentence sample, drawn randomly from each contrast condition. For each dataset and model, the highest surprisal value across conditions is underlined. The tested LLMs consistently match expectations from the literature, where **the more (psycho)linguistically complex condition has higher mean surprisal in every case** and statistical significance was determined by a one-sided t-test ($\alpha = 0.05$). In the **subordination/coordination** datasets, the *subordination* condition, given by a more nested syntactic structure, has higher surprisal than the *coordination* condition. For the **center/right branching** datasets, the *center embedding* condition has higher surprisal than the *right branching* case. Finally, for the **ambiguity** datasets, the *ambiguous attachment* condition has higher surprisal under an LLM than the unambiguous high and low attachment cases.

Higher-order linguistic processing around the ID peak Cheng et al. (2025) found that the ID peak span coincides with a phase in which higher-order linguistic information is made available, as indicated by a set of MLP-based probing tasks and downstream tests. We replicate their probing results for all tested models, using syntactic and semantic tasks from Conneau et al. (2018) (bigram shift, coordination inversion, and odd man out). While our results (in Fig. 5) are somewhat noisy, we confirm that performance on these tasks tends to grow to its maximum within the ID peak span. This confirms that the generic-sequence ID peak span provides a useful rule-of-thumb for the locus of deeper syntactic/semantic processing in the LLMs.

C.3 Intrinsic dimension profiles of the other models

ID profiles for the 3 models not shown in the main text (Gemma, Mistral and Qwen) are reported in Fig. 6.

C.4 Intrinsic dimension profiles for different levels of embedding and for *or*-coordination

In the main text, we present results for coordinated/subordinated sentences that contain 4 clauses. Fig. 7 compares these results to those obtained when using 3-clause or 2-clause sentences (e.g., respectively: "Quinn is rejoicing and/that Mary is screaming and/that the driver is faltering" and "Quinn is rejoicing and/that the driver is faltering"). We observe first of all that IDs, unsurprisingly, are higher for longer sentences. More importantly, the distinction between subordinated and coordinated sentences is consistent and strong in all models for the longest 4-clause sentences. It is present for most models also for 3-clause sentences, although typically with weaker separation. It is absent of very moderate for the shortest 2-clause sentences. It thus seems that the structural differences between flatter coordinated sentences and more hierarchical subordinated ones becomes salient enough to leave a recognizable trace in the ID profiles only above a certain degree of nesting.

The stability of our results concerning the subordination/coordination contrast is confirmed by Fig. 8, where, for the 4-clause case, we include another coordination dataset, namely one where the sentences are coordinated by *or* instead of *and* (see Table 2). *Or*-coordination displays lower ID than *that*-subordination, with a profile that is qualitatively similar to that of *and*-coordination. Interestingly, for all LLMs except Pythia, *or*-coordination has higher ID than *and*-coordination. We conjecture that this reflects higher semantic complexity, as *or* introduces a set of alternatives over possible states of the world, rather than merely conjoining statements about the same state of the world.

C.5 Intrinsic dimension for the subject vs. object relative clause contrast

Besides the processing differences depending on the positions of relative clauses (right branching vs. center embedding), we examined an additional condition based on embedded relative clauses. We

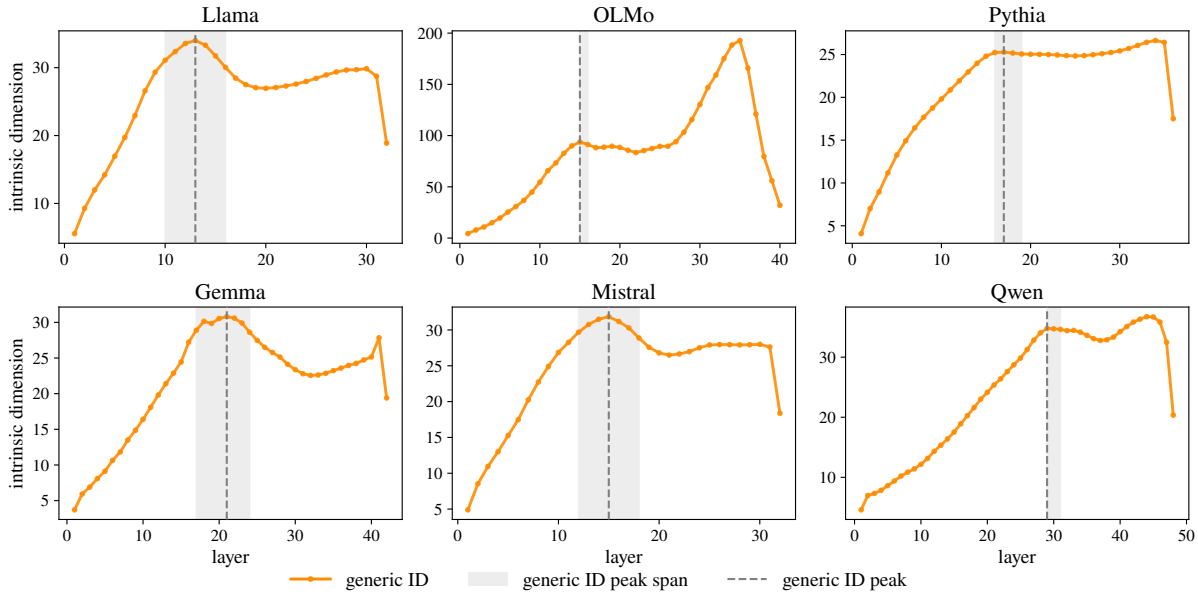


Figure 4: ID values across layers and estimated peak spans for all LLMs given an input of naturalistic corpus sequences. Mean values with (present but virtually invisible) error bars.

compared the ID of sentences with object relative clauses (ORC), corresponding exactly to the sentences in the center-embedding condition, and sentences with subject relative clauses (SRC); Table 3 reports examples of both. Compared to ORCs, SRCs are supposedly less complex, as they involve shorter syntactic dependencies and are aligned with the subject-verb-object order. This is reflected in their easier parsing by humans (Lau and Tanaka, 2021).

The ID patterns illustrated in Fig. 9 confirm this difference in complexity, showing consistently lower ID values across layers and models for SRCs. As in the right-branching/center-embedding contrast, the difference is often maximal in the pre-generic-ID layers. Interestingly, unlike right-branching structures, SRCs do not have high ID after the generic-ID peak. This supports our conjecture that the latter pattern in right-branching sentences is due to the fact that they end with an embedded clause (unlike center-embedded ORCs and SRCs). Another intriguing pattern is that SRCs (but not ORCs) have a clear peak coinciding with the generic one, so that the generic-ID-peak phase is actually the one where the difference between conditions is lowest.

C.6 Intrinsic dimension profiles for the three relative clause attachment conditions

In the main text and App. C.3, we compared the ambiguous attachment condition to relative clauses with unambiguously low attachment. Fig. 10 shows these two conditions together with a third, unambiguous *high* attachment condition. We see that there are only small and non-systematic differences between the two unambiguous attachment conditions.

C.7 Information Imbalance validation

To independently validate Information Imbalance as a measure of semantic information sharing, we created control contrasts between sentences expected to be nearly semantically equivalent, and between unrelated sentences. For the near-equivalent contrast, we chose the active/passive voice alternation. More precisely, we took the center-embedding dataset (third row of Table 1), and, for each sentence there, we created a variant with the relative clause turned into the passive voice. Thus, an example of the active/passive contrast is: “The waiter that Bill humiliated was snickering” vs. “The waiter that was humiliated by Bill was snickering”. The corresponding Information Imbalance profiles are shown in Fig. 11. As expected, Information Imbalance for these close paraphrases is consistently extremely low, approaching 0. Further, as expected for near-equivalent sentences, the Information Im-

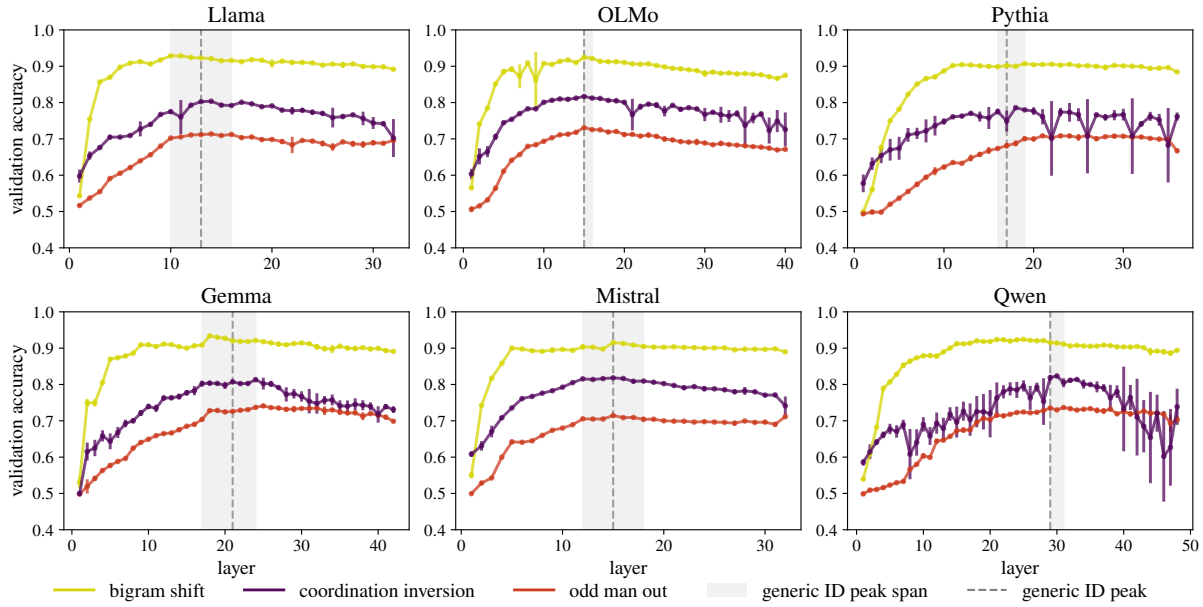


Figure 5: Accuracies in three probing tasks (bigram shift, coordination inversion and odd man out) across layers, for all models. Means and standard errors across five seeds.

balance in the two directions is perfectly symmetrical. There is a very slight increase in Information Imbalance towards the final layers, which might be related to how the information structures associated to active vs. passive might favor different continuations.

We then used the same datasets, but now *shuffling* the order of the passive sentences, so that each original active form would be paired with the passive of another active form. For example, "The waiter that Bill humiliated was snickering" might be paired with "The potters that were advised by the politicians were waiting". Fig. 12 shows that, for these unrelated sentences, Information Imbalance is consistently around 1, corresponding to no information sharing between the pairs (Glielmo et al., 2022).

The comparison between the active/passive and shuffled contrasts thus validates Information Imbalance as a way to capture semantic similarity between sentences. We point to Acevedo et al. (2026) for futher validation and theoretical comparisons between Information Imbalance and alternative information sharing measures.

C.8 Information Imbalance profiles of the other models

Fig. 13 reports the Information Imbalance (Δ) results for the 3 models not shown in the main text (Gemma, Mistral and Qwen), confirming the trends we saw there.

C.9 Pruning profiles of the other models and contrasts

Pruning results with the coordinated/subordinated datasets for the 3 models not shown in the main text (Gemma, Mistral and Qwen) are reported in Fig. 14. The different responses to pruning are more visible in early layers and a dip in accuracy specifically affecting the coordinated set is found in correspondence to the ID peak in Mistral, similar to Llama, OLMo and Pythia.

Pruning results for the other contrasts are reported in figures 15 and 16, with differences across conditions scarcely observable.

C.10 Pruning profiles with KL divergence

Fig. 17 shows pruning profiles over layers for the crucial coordination/subordination contrast, where pruning impact is measured using KL divergence (Kullback and Leibler, 1951), a continuous way to measure the difference between the next-token probability distribution before and after layer pruning. This analysis yields similar qualitative takeaways to those obtained through next-token prediction pruning accuracy. In particular, we observe upwards spikes (*higher* KL means higher pruning impact) in the present plots that are aligned with the downwards dips (*lower* accuracy means higher pruning impact) in the pruning accuracy plots of

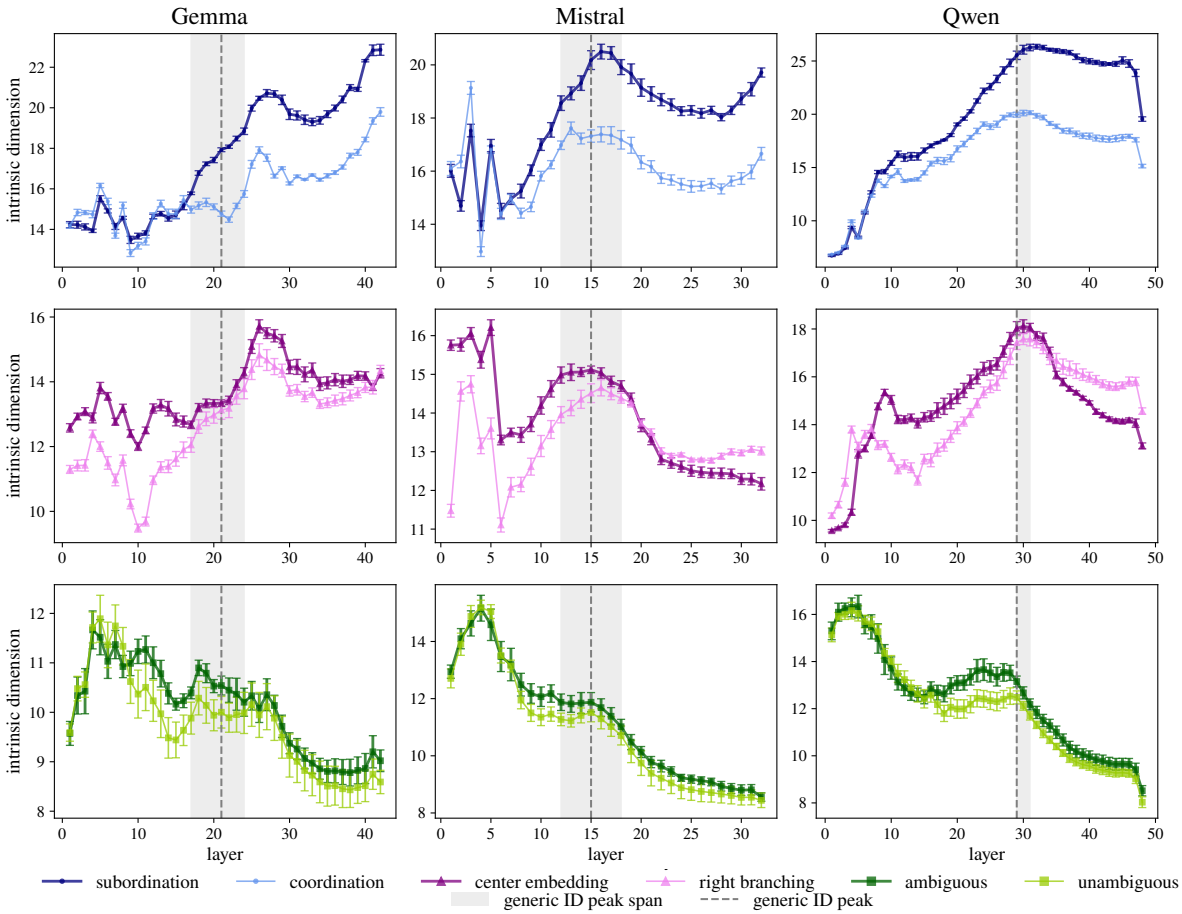


Figure 6: ID profiles through LLM layers (means and error bars across 5 partitions). Vertical dashed line marks maximum ID on generic sequences, and shaded area the corresponding span, estimated as explained in App. C.2.

Figures 3 and 14, in coincidence with the generic-ID peaks.

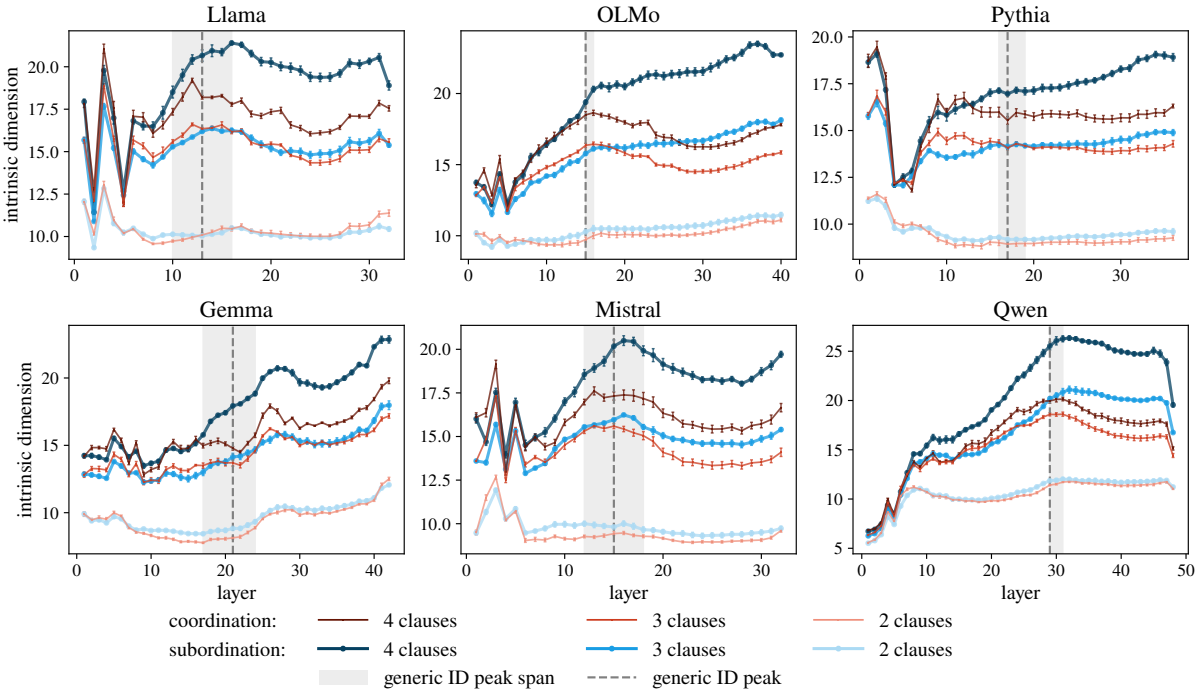


Figure 7: ID profiles through LLM layers (means and error bars across 5 partitions) for coordination/subordination sets featuring different numbers of clauses. Vertical dashed line marks maximum ID on generic sequences, and shaded area the corresponding span.

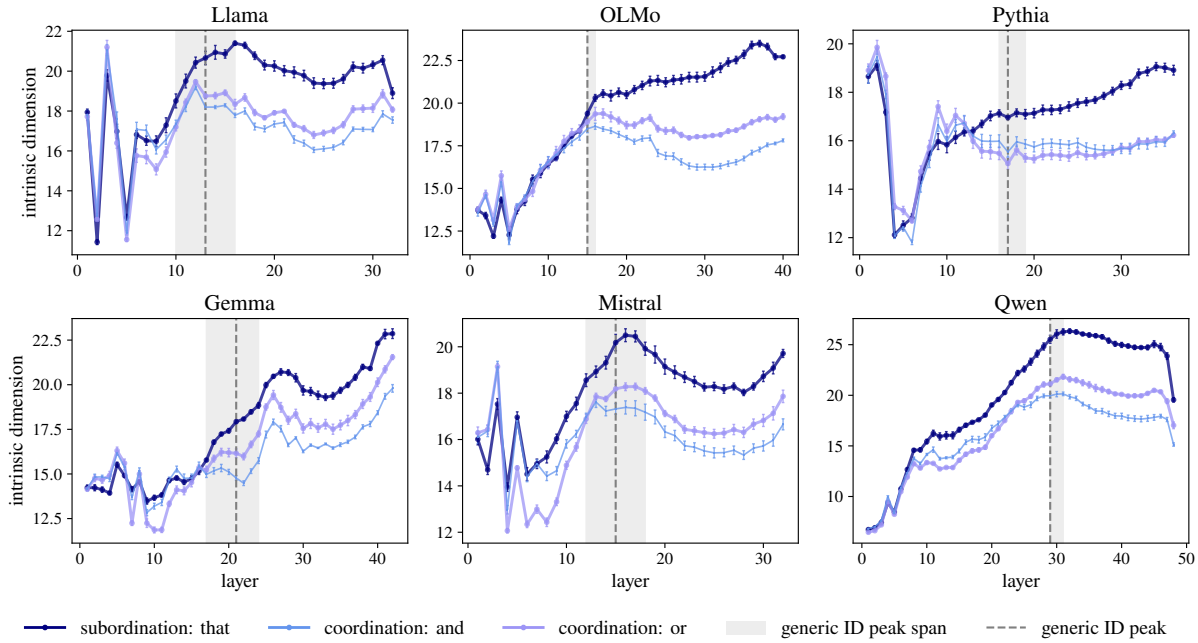


Figure 8: ID profiles through LLM layers (means and error bars across 5 partitions) for subordination vs. coordination, including subordinated constructions using *or*. Vertical dashed line marks maximum ID on generic sequences, and shaded area the corresponding span.

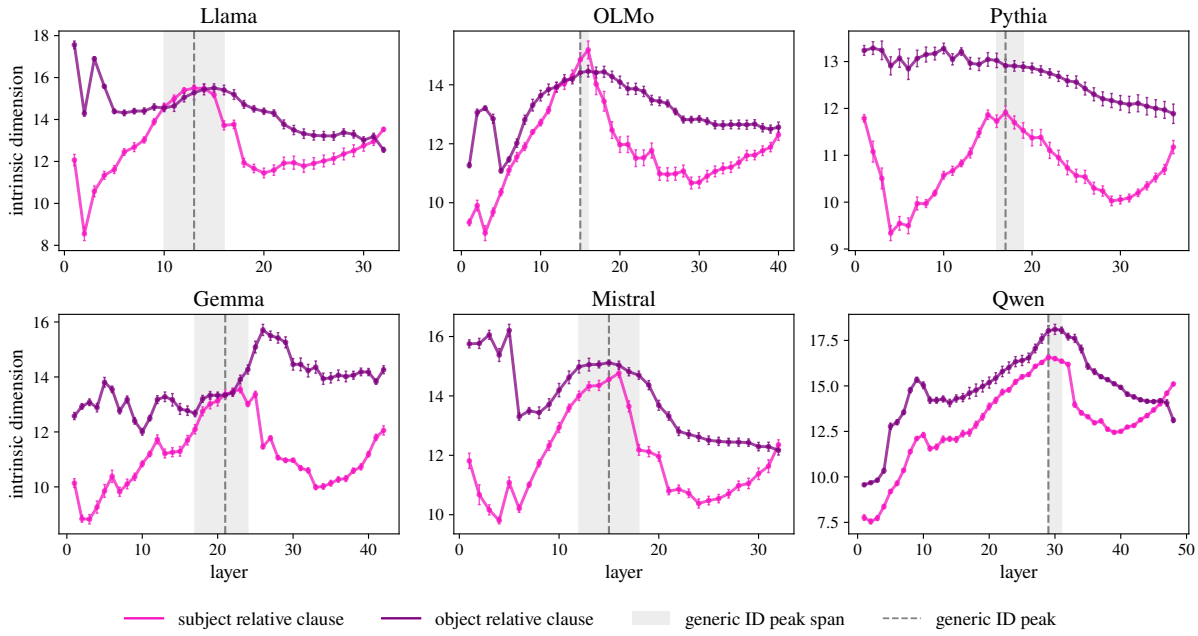


Figure 9: ID profiles through LLM layers (means and error bars across 5 partitions) for the subject relative clause/object relative clause contrast.

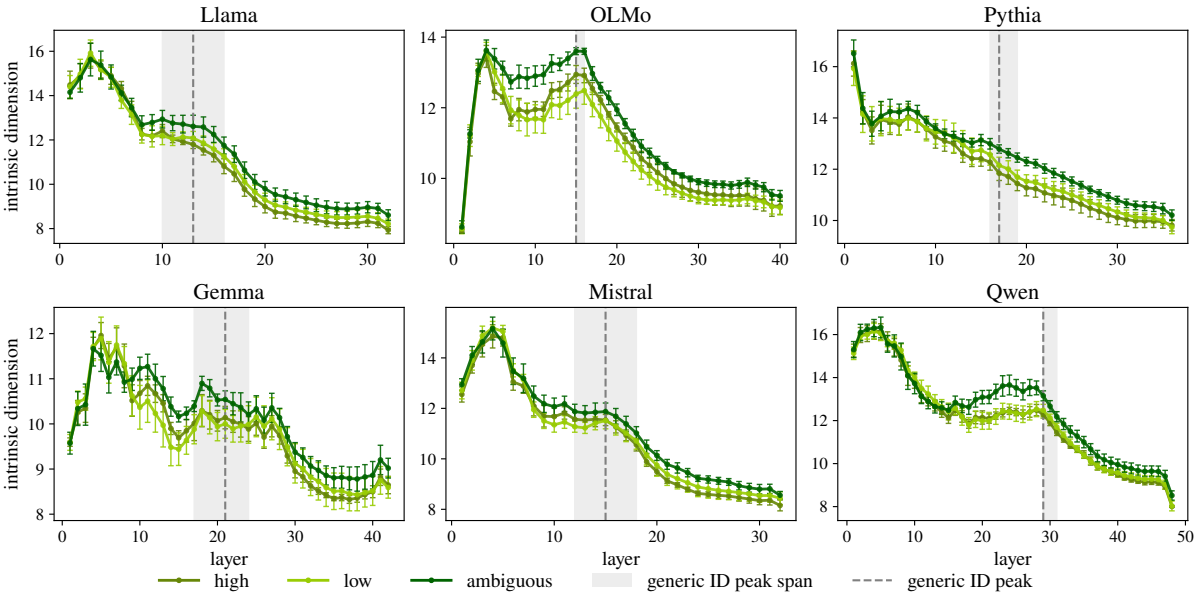


Figure 10: ID profiles through LLM layers (means and error bars across 5 partitions) for the high, low and ambiguous relative clause attachment conditions. Note that low attachment corresponds to the unambiguous condition shown in the last rows of figures 1 and 6. Vertical dashed line marks maximum ID on generic sequences, and shaded area the corresponding span.

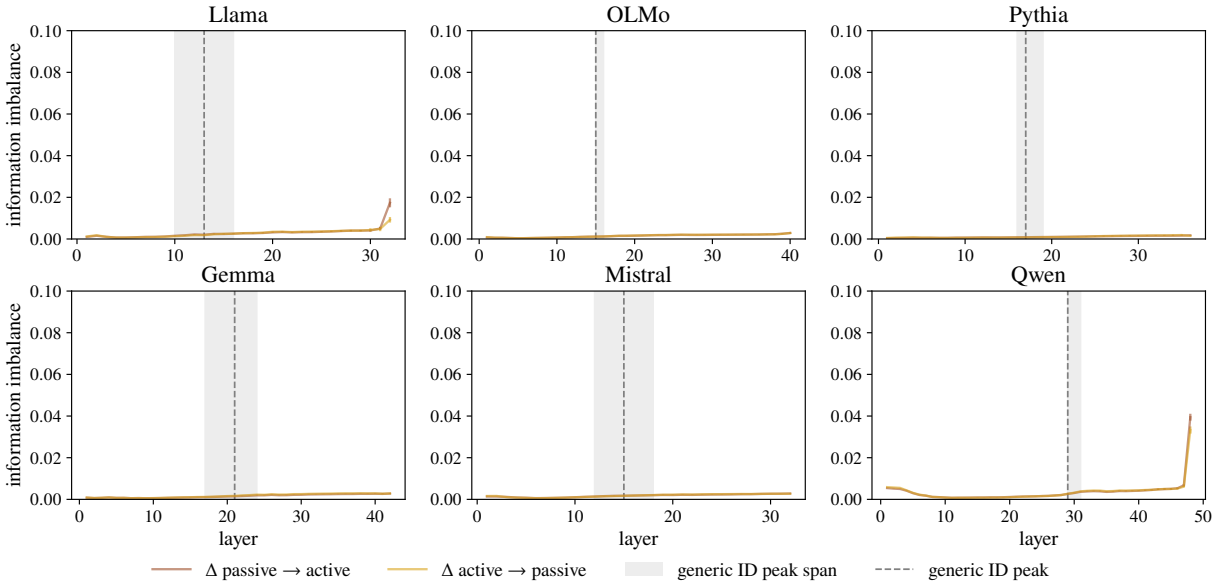


Figure 11: Information Imbalance Δ between active and passive sentences for all models: means across 5 partitions with (present, but virtually invisible) error bars. Shaded area marks generic ID-peak span, with a vertical dashed line at the generic-ID maximum. Higher Δ means lower similarity.

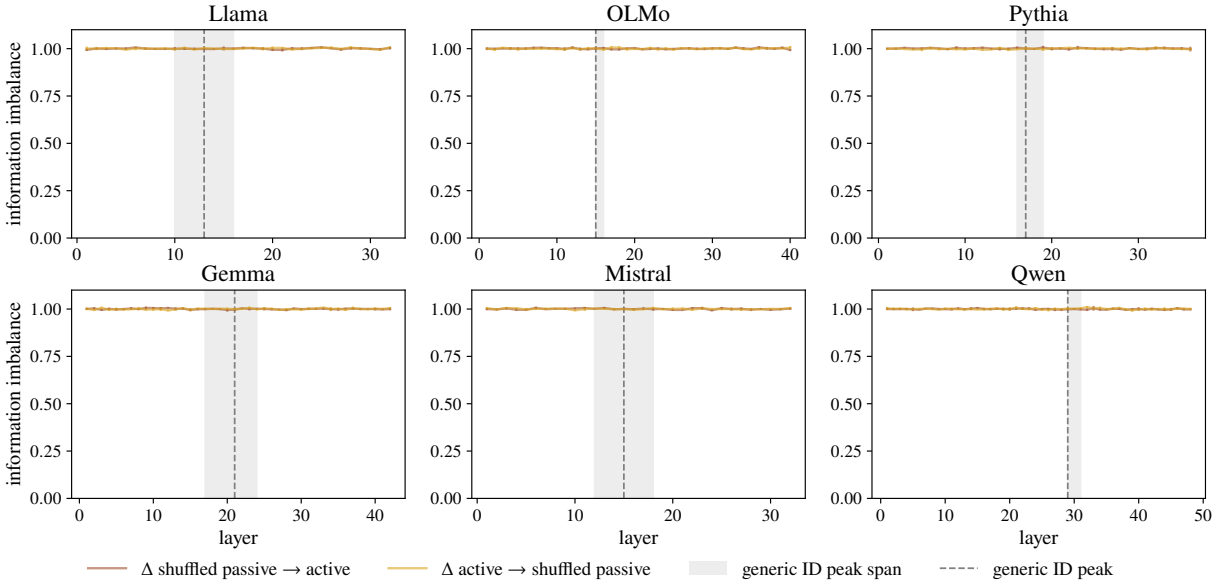


Figure 12: Information Imbalance Δ between shuffled active and passive sentences for all models: means across 5 partitions with (barely visible) error bars. Shaded area marks generic ID-peak span, with a vertical dashed line at the generic-ID maximum. Higher Δ means lower similarity.

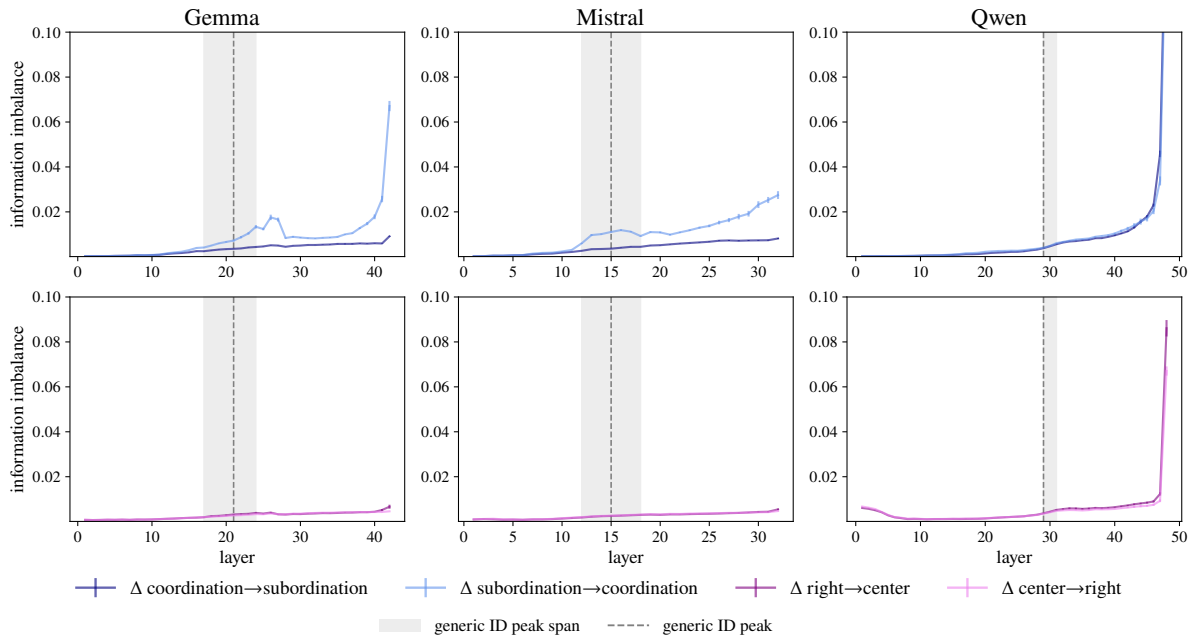


Figure 13: Information Imbalance Δ between coordinated/subordinated sentences (top) and right-branching/center-embedding sentences (bottom): means across 5 partitions with error bars (often so small as to be invisible), for the Gemma, Mistral and Qwen models. Shaded area marks generic ID-peak span, with a vertical dashed line at the generic-ID maximum. Higher Δ means lower similarity.

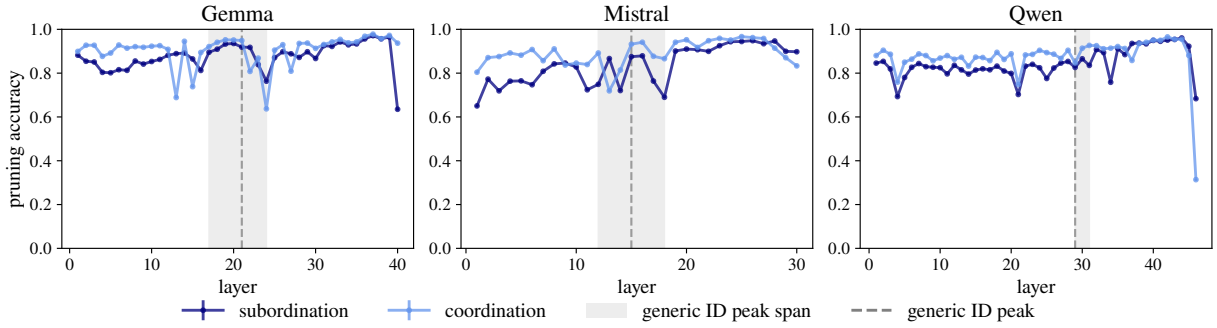


Figure 14: Pruning accuracies comparing the coordination vs. subordination conditions for Gemma, Mistral and Qwen. Means and (present but virtually invisible) standard error bars over 5 partitions.

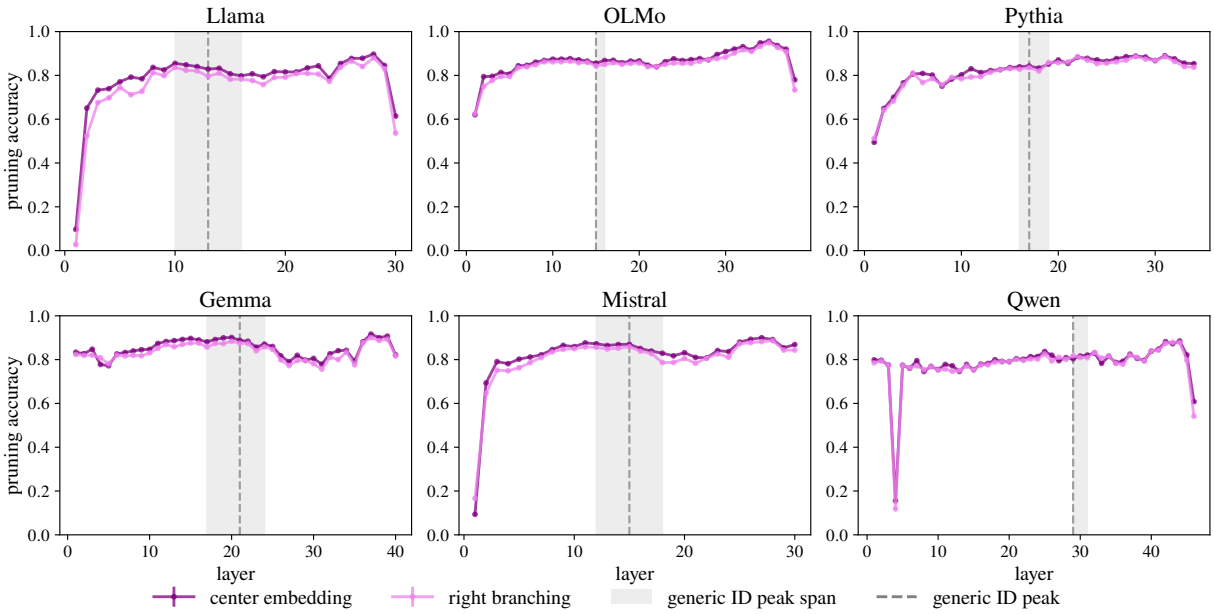


Figure 15: Pruning accuracy for right branching vs. center embedding. Means and (hardly visible) standard errors over 5 partitions

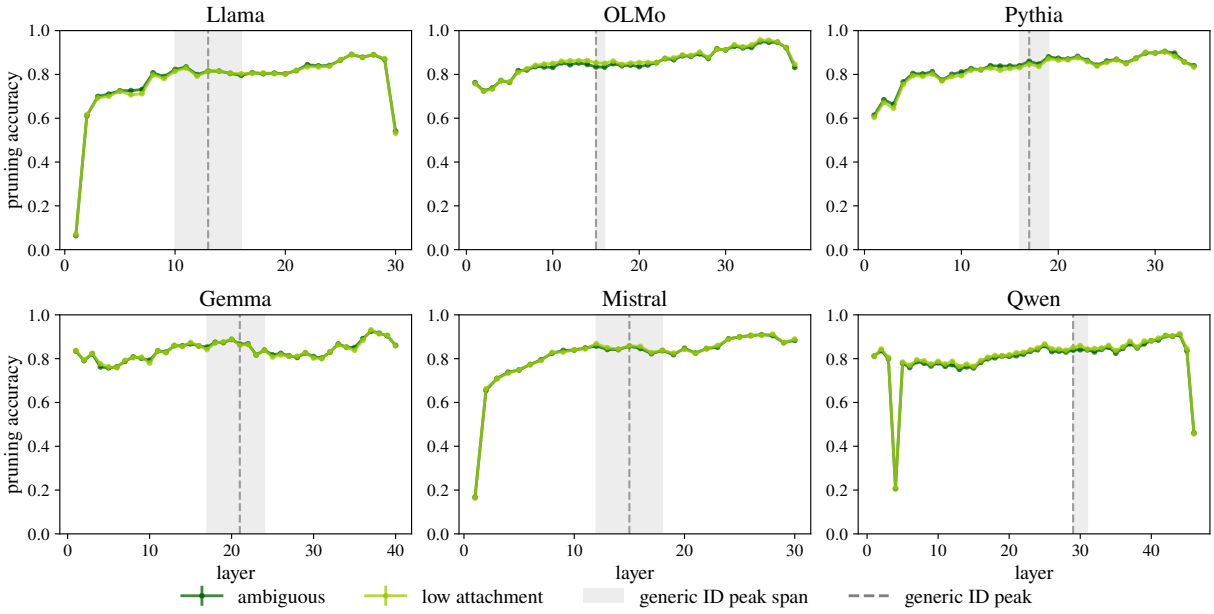


Figure 16: Pruning accuracy for the ambiguous vs. unambiguous conditions. Means and (hardly visible) standard errors over 5 partitions.

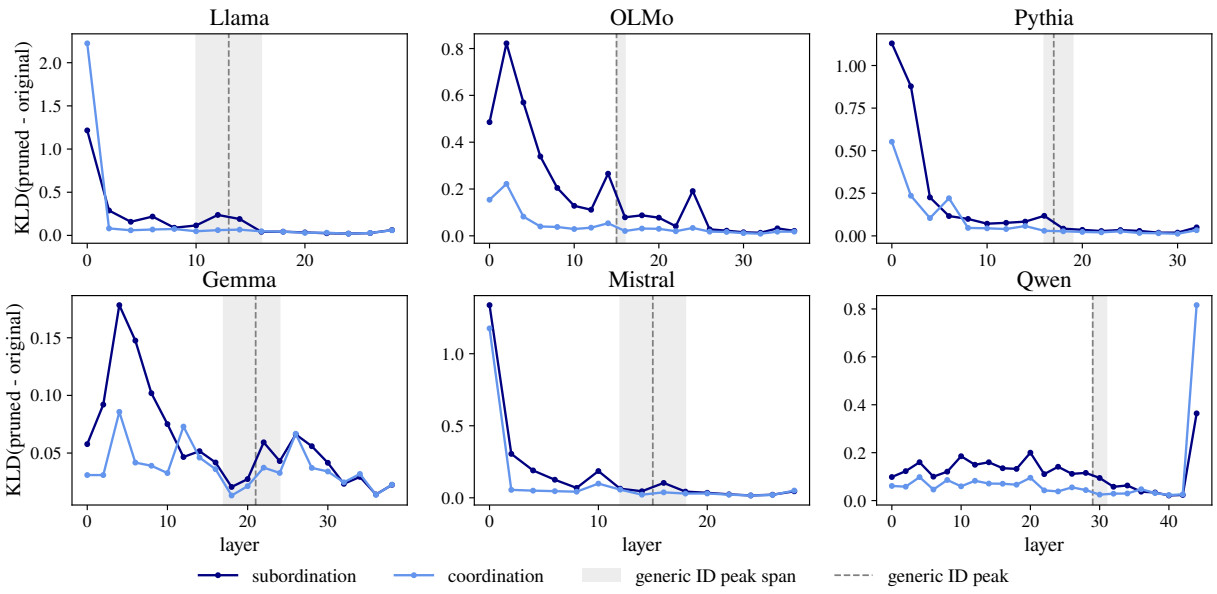


Figure 17: Pruning effects measured through the KL divergence from the pruned to the original prediction, every two layers. For each model, the subordination (dark blue) and coordination (light blue) KL are shown against the pruned layer index. All curves show the mean over 5 random partitions, where one standard error is shaded (though very small, thus not visible in the plots).